

KANSAS STATE
College of Engineering

**IMSE 662 COMPUTER AIDED
MANUFACTURING
FINAL REPORT**



EXECUTIVE SUMMARY

For the final project the team was tasked with designing and implementing a computer-controlled manufacturing system, utilizing different equipment and systems that were introduced in lecture. The equipment used was an Allen-Bradley Micro850 PLC, a Mitsubishi Movemaster EX robotic arm, a CNC laser router, and a Shuttleworth roller conveyer. In this project, the team gained a more complete understanding of PLC ladder logic, robotic work envelopes, robot programming, and programming using G-Code. After completing this project, the team realized how complicated it is to create, integrate, and optimize a manufacturing setting with multiple different operations.

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INTRODUCTION

In this project, teams were tasked with creating, designing, and implementing a computer-controlled manufacturing system utilizing the equipment in the laboratory. The equipment included an Allen-Bradley Micro850 PLC, a Mitsubishi Robot Movemaster EX, a laser router, and a Shuttleworth roller conveyor system. Teams were tasked with designing the entire manufacturing process from start to finish, transporting a wooden slab down the conveyor, transported by the robot arm onto a table to be engraved by a laser. The team 3D printed a fixture to hold the slab, after not finding enough stability from many designs on the given pegboard. After the fixture was installed, the team ran several successful tests of the whole manufacturing process.

INSTRUMENTS & COMPONENTS

- Allen-Bradley Micro850 PLC
- Mitsubishi Movemaster EX
- Laser router
- Shuttleworth roller conveyor
- 3D printed fixture

PROCEDURE

CONVEYER

1. Define input and output variables
2. Define how inputs interact with outputs
3. Consider how inputs and outputs are called (rising edge, trailing edge, pulse, constant logic which will determine coil type, direct/set)
4. Create Ladder diagram
 - a. Optionally, create smaller ladder diagrams to test individual scenarios
5. Test with an arbitrary robot-engraver delay
6. Adjust accordingly
7. Integrate

ROBOT ARM

1. Plan path
 - a. Plan the most efficient route that the robot arm can take without crashing into any stationary objects.
2. Create raft work holding
3. Define positions by manually jogging the robot arm using the teaching box

4. Run through positions according to planned path to ensure clearance
5. Write program
6. Run program and adjust timings for gripper actuation and CNC Router as necessary

LASER ROUTER

1. Create sketch to be laser engraved with dimensions
2. Write program
 - a. Set unit type
 - b. Set local origin
 - c. Translate sketch to G-Code
 - d. Return to local origin
 - e. Clear workpiece
3. Integrate

DISCUSSION

CONVEYER

The Shuttleworth roller conveyer system had two levels accessed by lifting conveyors N and S, additionally it had various proximity sensors, gates, and a pusher. We defined the inputs as the 13 active sensors and the outputs as gates A, B, C, and E, pusher, and lifting conveyor N and S. Initially, smaller ladder diagrams were used to understand how the outputs were activated. We found that gate A had a hardware issue, certain gates activated opposite to others. Then, we proceeded to determine how each individual sensor should affect outputs in front and behind the sensor. For instance, when the raft travels over sensor 2, the S lifting conveyor should lift up and move west such that the raft continues until sensor 3 is activated to stop and lower the conveyor where the raft accesses the main loop. Once the ladder diagram was complete, we tested the code with a small arbitrary delay for the workpiece pickup region and noticed certain parts were actuating too quickly causing the raft to rotate or crash. These issues were resolved by adding short delays to gates and lowering and stopping the lifting conveyors. After some iteration, we deemed the conveyor was completed and would wait for system integration tests to determine the final delay for the robot arm and laser router.

LADDER DIAGRAM

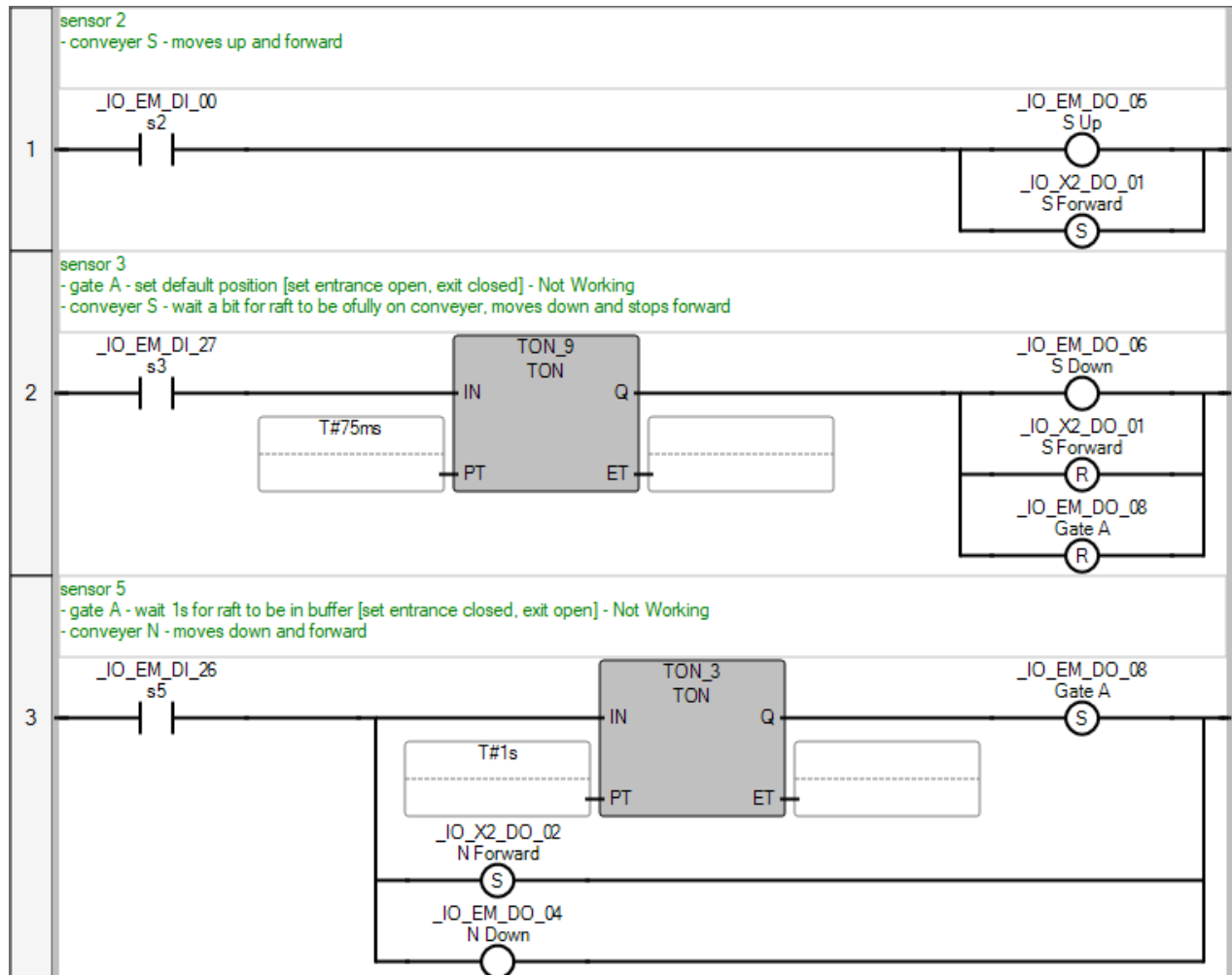


Figure 1: Ladder Diagram Pt. 1

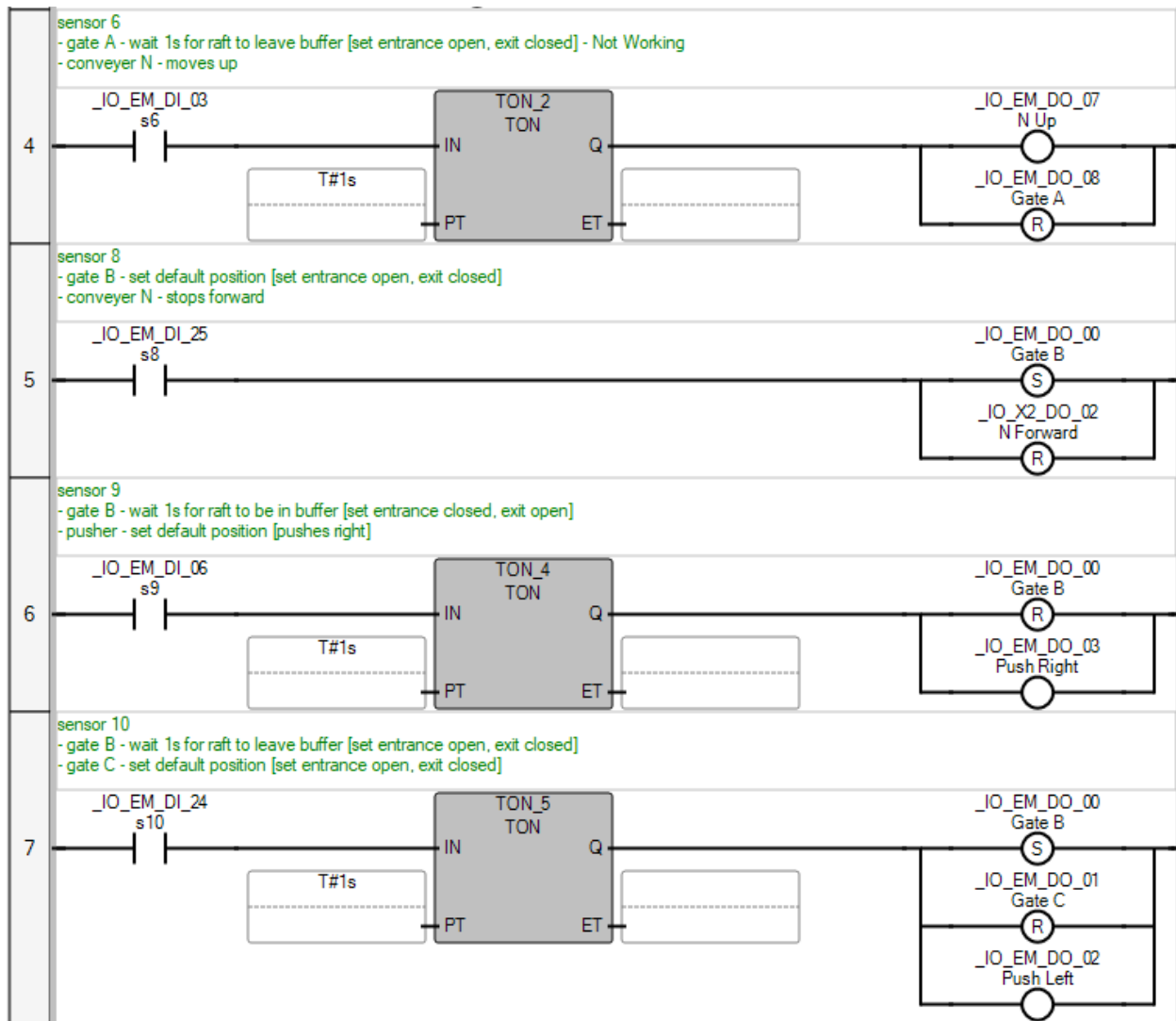


Figure 2: Ladder Diagram Pt. 2

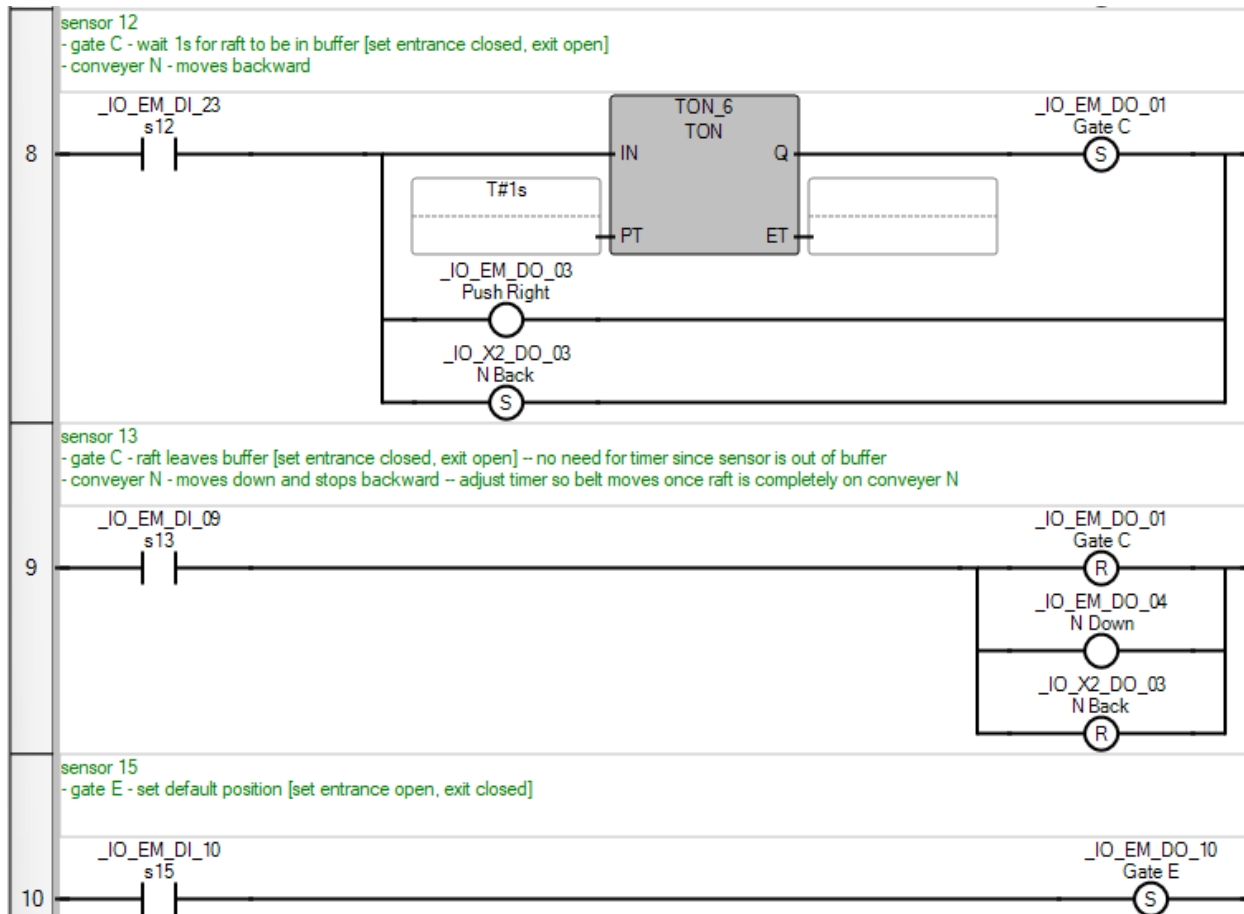


Figure 3: Ladder Diagram Pt. 3

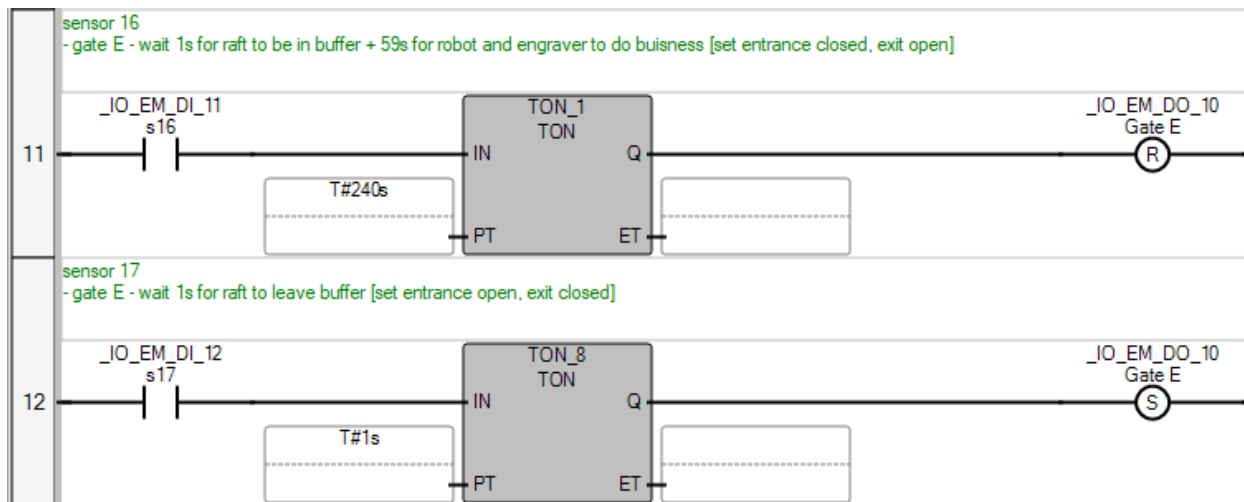


Figure 4: Ladder Diagram Pt. 4

The ladder diagram was the first piece of the puzzle that we figured out. Using our knowledge of *Set* and *Reset* coils, and our new practice with *TON* timers, we programmed the ladder diagram together in the most efficient way we could. The first steps of the process were defining what our

input variables were for the program to read. We then went to each sensor step by step, checking what pieces of machinery we needed to raise, lower, run forward or backward, and when to signal the gates to swap positions so our raft could continue through the system.

Making this assembly line into a ladder diagram was not without its challenges, though. One of the first problems we ran into was the gates themselves. We did not know that the gates were set to 'Up' as their default position, so when we tried using a direct coil, we thought it would pulse the gate enough for it to set to its 'Down' position. We quickly found out how wrong that was when we ran the code for the first time, and the gates did not move at all. To fix this issue, we switched to Set and Reset coils, which proved to be successful in lowering the gates. From there, it was a relatively short process to set the correct delay on each TON timer in the system to wait for the robot and laser engraver to complete their processes. In the end, the whole team learned a lot about the details of ladder diagrams and how useful they are in a manufacturing setting.

ROBOT ARM

The first step of programming the robot arm is to plan out the path the robot will take as it delivers the workpiece from the raft to the CNC router worksurface. The main considerations taken were to avoid crashes with stationary objects, provide repeatable locating of the part on the worksurface, and to take paths that would not crash if the robot overshot its target. The wooden workpiece had to be held in a repeatable position upon the raft, to allow the robot to accurately retrieve and return the workpiece. Originally, we were going to use the provided dowels and rubber bands to hold the workpiece. However, we quickly found that the workpiece would slide around more than we deemed allowable. To combat this issue, we decided to 3-D print a custom fixture that held the workpiece.



Figure 5: Image of 3D printed raft

Then we had to manually jog the robot arm to the predetermined position using the teaching box. These positions can be moved to in either a Cartesian or articulating fashion. When the desired position for a given step is reached, the "P.S.", "###", and "ENT" keys are pressed successively

to store that position in the robot's memory. The “###” represents the position number that will be associated with the position. In our case, we were assigned the position numbers from 100-199.

After storing all the positions that were planned in the first step, we ran through all the positions in the order that will be used in the final program to ensure proper clearance throughout the entire program. Next, we wrote the program using various *move*, *wait* and *speed change* commands, with respect to how much time it would take to actuate the gripper and how long the CNC Router operation took.

ROBOT CODE

Table 1: Robot Arm Code

Command	Description
TI 50	Wait 5 seconds
NT	Move to Nest
SP 5	Set speed to 5
MO 105	Move to position 105
SP 2	Set speed to 2
MO 100	Move to position 100
TI 50	Wait 5 seconds to actuate gripper
MO 110	Move to position 110
SP 5	Set speed to 5
MO 120	Move to position 120
MO 130	Move to position 130
MO 140	Move to position 140
MO 150	Move to position 150
SP 2	Set speed to 2
MO 160	Move to position 160
TI 50	Wait 5 seconds to actuate gripper
MO 150	Move to position 150
TI 900	Wait 90 seconds for laser operation
MO 160	Move to position 160
TI 50	Wait 5 seconds to actuate gripper
MO 150	Move to position 150
SP 5	Set speed to 5
MO 140	Move to position 140
MO 130	Move to position 130
MO 120	Move to position 120
MO 110	Move to position 110
MO 100	Move to position 100
TI 50	Wait 5 seconds to actuate gripper
MO 105	Move to position 105
NT	Move to Nest

LASER ROUTER

For the laser router, we were tasked to route our initials into the work pieces. SolidWorks was used to create a 2D sketch within the boundary of the workpiece, along with dimensions from where the local origin is to each point on the sketch. This was done to easily translate and program the laser router to move along the sketch. “G21” was used first to denote the units in millimeters. Following, “G0” quickly moves the router linearly to a safe moving height then moves to the rough local origin using “G52”. The translated sketch follows in blocks corresponding to each letter. “M3 S500” turns the laser on at power 500, “G1” moves the laser slowly, and “M5” turns the laser off. Additionally for the arcs, “G2” and “G3” are used by rotating from the center of the arc clockwise and counterclockwise, respectively, denoted by “I”, “J”, and “K”. Lastly, after the complete sketch, the laser returns to the local origin and moves the gantry where the laser resides on away from the workpiece to allow it to be picked up by the robot arm. System integration tests would finalize the local origin and provide the time it takes to complete this part of the system.

G-CODE

Table 2: Laser Engraver G-Code

G21	; Letter I Start
G0 Z50	G0 X35 Y45
	M3 S500
	G1 X55 Y45 F500
	M5
; Set Local Origin	G0 X45 Y45
G52 X298 Y93	M3 S500
G0 X0 Y0	G1 X45 Y20 F500
	M5
; Letter W Start	G0 X35 Y20
G0 X35 Y80 Z30	M3 S500
M3 S500	G1 X55 Y20 F500
G1 X35 Y55 F500	M5
G1 X45 Y67.5	; Letter I End
G1 X55 Y55	
G1 X55 Y80	
M5	; Letter B Start
; Letter W End	G0 X98.75 Y45
	M3 S500
	G1 X105 Y45
; Letter S Start	G2 X105 Y32.5 J-6.25
G0 X98.75 Y61.25	G1 X98.75 Y32.5
M3 S500	G1 X105 Y32.5
G3 X105 Y67.5 I6.25 F500	G2 X105 Y20 J-6.25
G2 X111.25 Y73.75 J6.25	G1 X98.75 Y20
M5	G1 X98.75 Y45
; Letter S End	M5
; Continued on next column	

	; Letter B End ; Return to Local Origin G0 X0 Y0 Z50 ; Move gantry away for robot arm G0 X-100 Y0
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After each component was completed, we integrated and tested the system as a whole with the laser powered off, allowing us to adjust delays/timers/pauses and positioning in each component. This provided us a complete system-wide travel time for a workpiece of about 5 minutes.

CONCLUSION

In conclusion, our team successfully designed, integrated, and tested the entire manufacturing process for our slab of wood. The wood was transported down a line of conveyors, which trigger sensors programmed in our ladder diagram to signal the lifts, pushes, and belts in the system. The raft reached the end of the line, hitting the last gate and setting a four minute timer. Our programming for the robot arm then came into effect, picking up the slab and transporting it safely onto the engraver table. Our laser engraver then exacted our G-Code written in an earlier step, engraving our first initials (WSIB) into the slab. After removing the safety barriers from the engraver, the robot automatically moves into place, grabbing the processed slab and placing it back down into our fixture on the table. With several seconds to spare, the last gate drops, sending our processed slab out to the shipping department. Through this process, the team learned how to design a ladder diagram using set, reset, and direct coils, as well as timers. The team also learned how to program a robot arm by setting specific positions in space, giving a smooth and safe path for the robot to follow. The team also learned how to write G-Code manually, programming a laser engraver to score four letters into the wooden slab. All in all, our team demonstrated comprehensive knowledge of small manufacturing processes, pushing through challenges and technical difficulties to make our final demo run smoothly.

APPENDIX

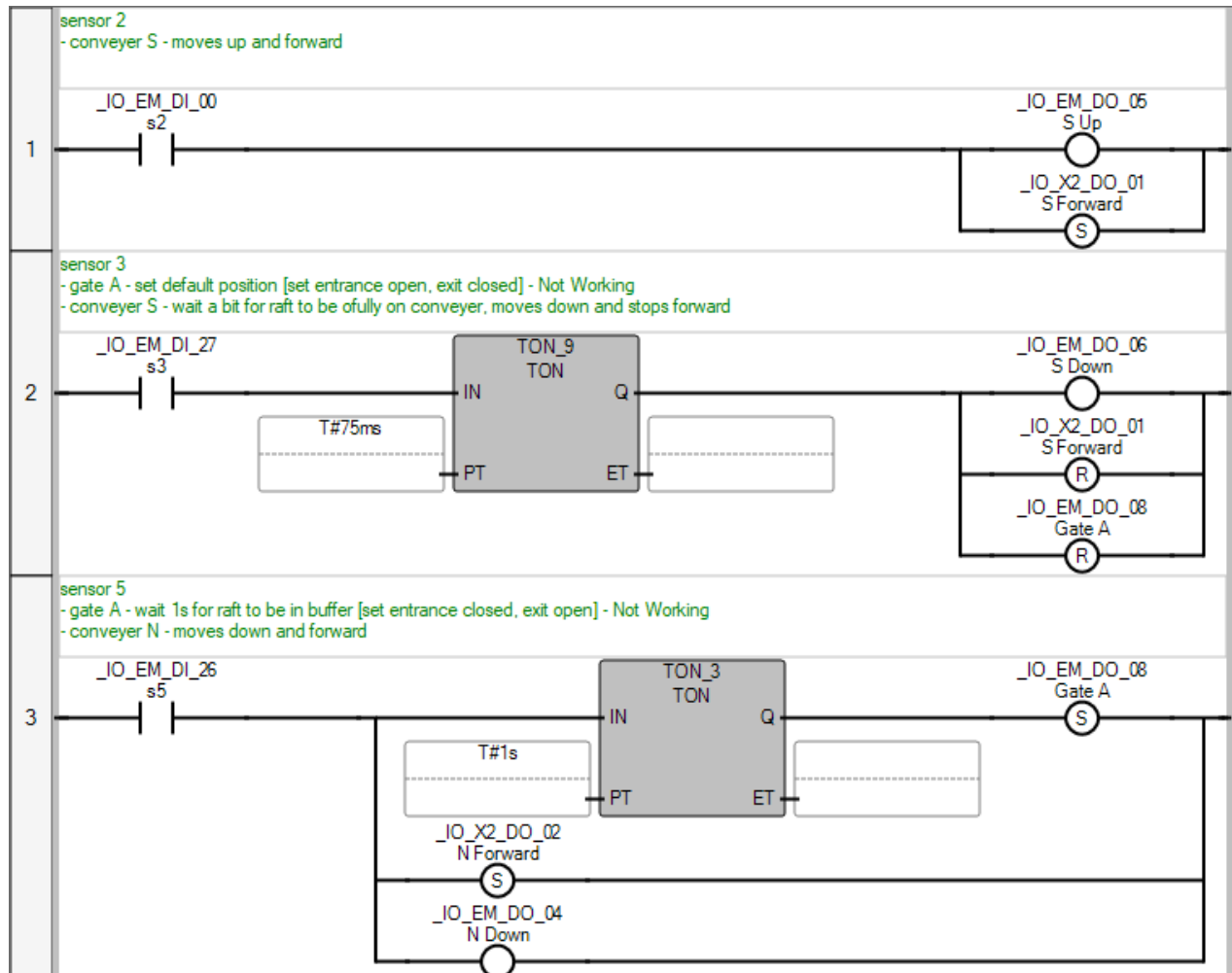


Figure 6: Ladder Diagram Pt. 1

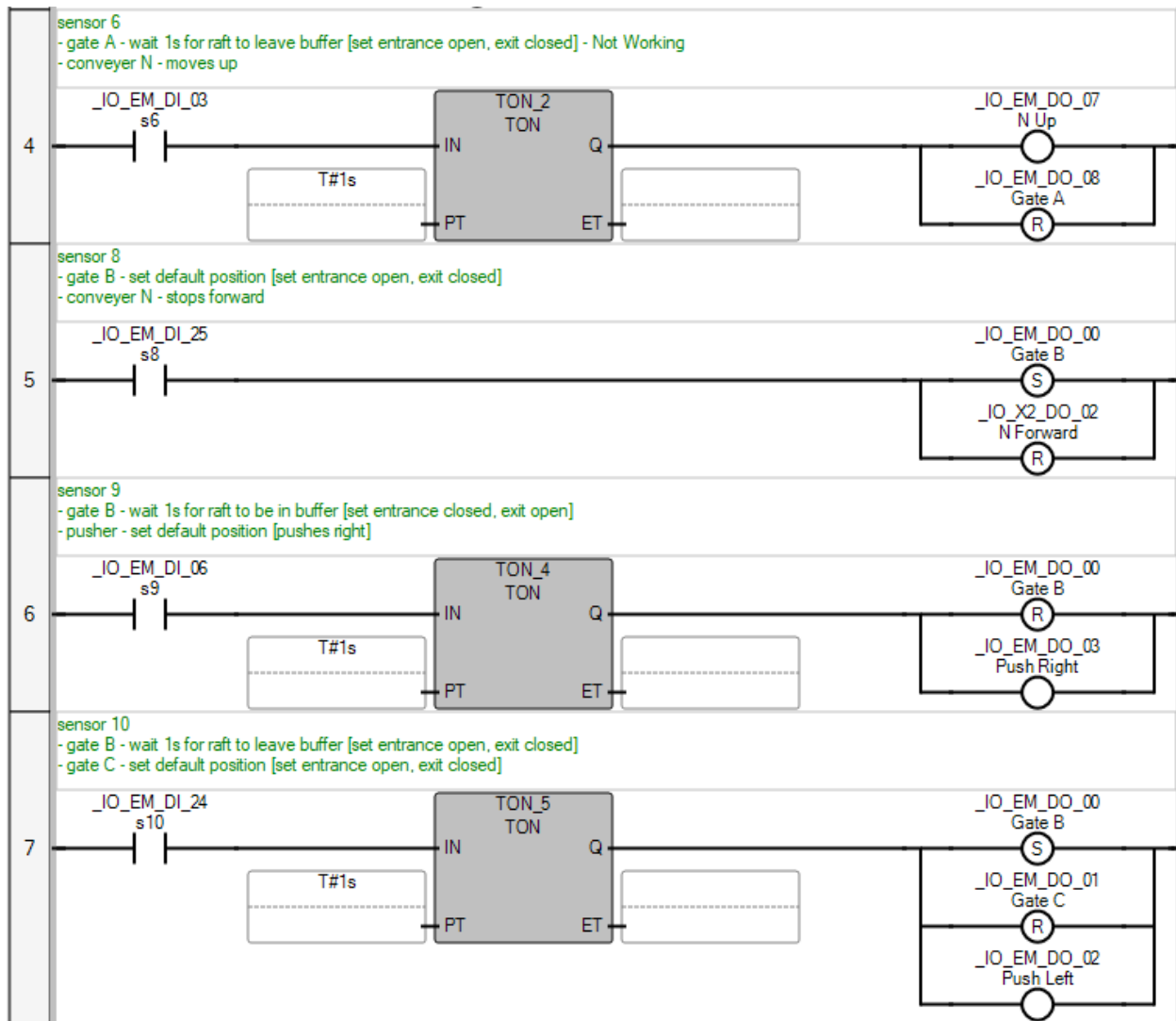


Figure 7: Ladder Diagram Pt. 2

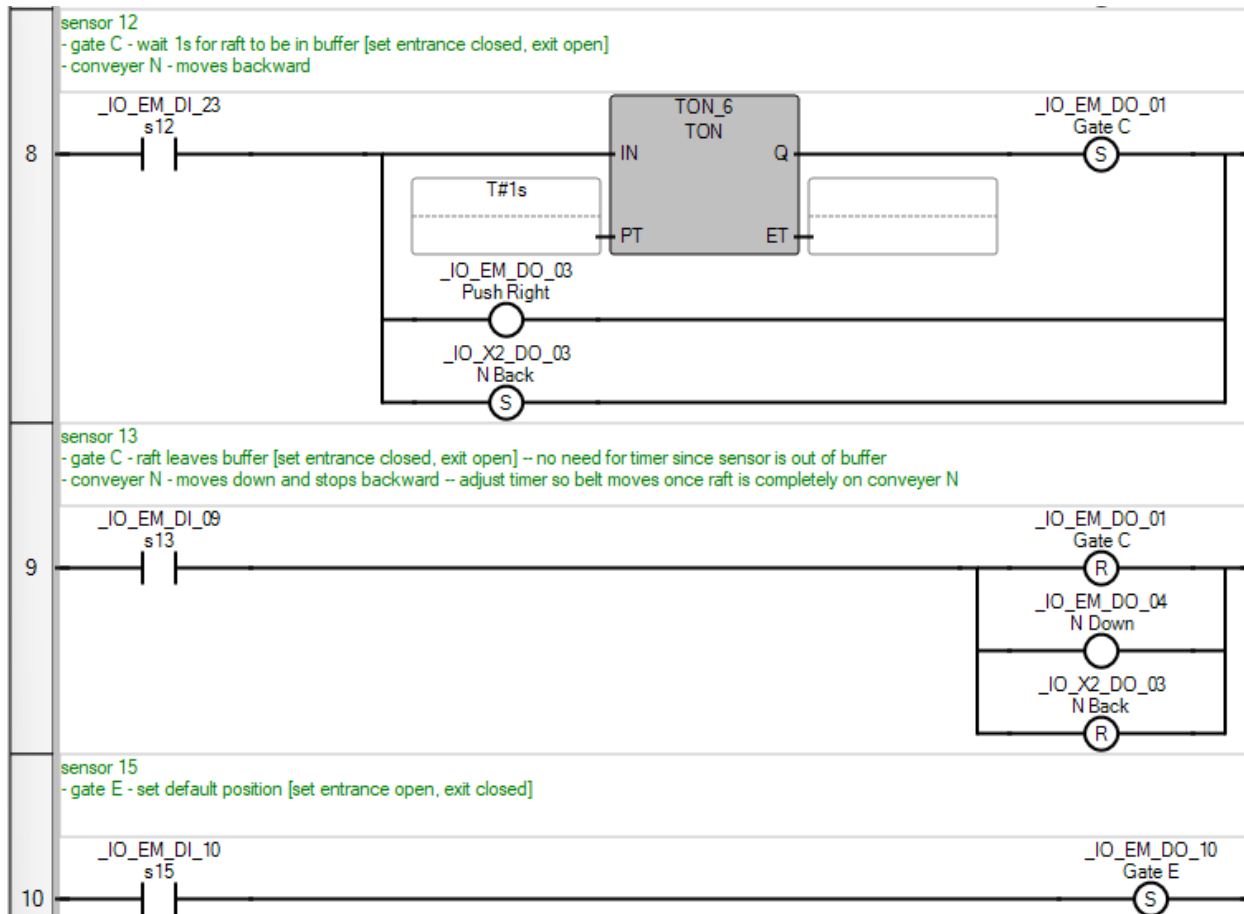


Figure 8: Ladder Diagram Pt. 3

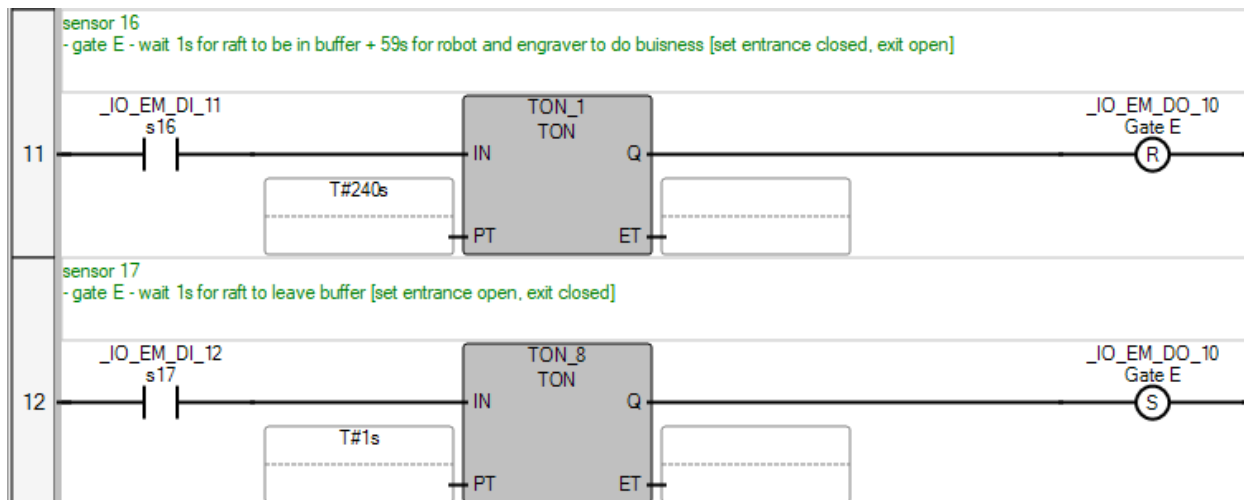


Figure 9: Ladder Diagram Pt. 4



Figure 10: Image of 3D printed raft

Table 3: Robot Arm Code

Command	Description
TI 50	Wait 5 seconds
NT	Move to Nest
SP 5	Set speed to 5
MO 105	Move to position 105
SP 2	Set speed to 2
MO 100	Move to position 100
TI 50	Wait 5 seconds to actuate gripper
MO 110	Move to position 110
SP 5	Set speed to 5
MO 120	Move to position 120
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MO 150	Move to position 150
SP 5	Set speed to 5
MO 140	Move to position 140
MO 130	Move to position 130
MO 120	Move to position 120
MO 110	Move to position 110
MO 100	Move to position 100

TI 50 MO 105 NT	Wait 5 seconds to actuate gripper Move to position 105 Move to Nest
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Table 4: Laser Engraver G-Code

G21 G0 Z50 ; Set Local Origin G52 X298 Y93 G0 X0 Y0 ; Letter W Start G0 X35 Y80 Z30 M3 S500 G1 X35 Y55 F500 G1 X45 Y67.5 G1 X55 Y55 G1 X55 Y80 M5 ; Letter W End ; Letter S Start G0 X98.75 Y61.25 M3 S500 G3 X105 Y67.5 I6.25 F500 G2 X111.25 Y73.75 J6.25 M5 ; Letter S End ; Continued on next column	; Letter I Start G0 X35 Y45 M3 S500 G1 X55 Y45 F500 M5 G0 X45 Y45 M3 S500 G1 X45 Y20 F500 M5 G0 X35 Y20 M3 S500 G1 X55 Y20 F500 M5 ; Letter I End ; Letter B Start G0 X98.75 Y45 M3 S500 G1 X105 Y45 G2 X105 Y32.5 J-6.25 G1 X98.75 Y32.5 G1 X105 Y32.5 G2 X105 Y20 J-6.25 G1 X98.75 Y20 G1 X98.75 Y45 M5 ; Letter B End ; Return to Local Origin G0 X0 Y0 Z50 ; Move gantry away for robot arm G0 X-100 Y0
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